(a) Training phase

Final Decoder Feature
$\boldsymbol{F} \in R^{C \times H \times W}$

Head

Segmentation Head
Logits: $\hat{Y} \in R^{C \times H \times W}$

Segmentation Loss
(Soft CE + Dice)

BD Head
$U = \mathrm{ReLU}(Conv_{k \times k}(F))$
$\boldsymbol{B}$:$1 \times H \times W$, $\boldsymbol{D}$:$1 \times H \times W$

Boundary Loss
$\lambda_{bd} \cdot BCE(B, B^*)$

Distance Loss
$\lambda_{dist} \cdot SmoothL1(D, D^*)$

Joint Training Loss
$L_{joint} = L_{seg} + \lambda_{bd} \cdot L_{BCE} + \lambda_{dist} \cdot L_{SmoothL1}$

(b) Inference phase

Final Decoder Feature
$\boldsymbol{F} \in R^{C \times H \times W}$

Segmentation Head
Logits: $\hat{Y} \in R^{C \times H \times W}$

Softmax over C classes

Argmax (per pixel)

Final Segmentation Mask